

LED Power Consumption

How the EHS Production Tool calculates LED wall power & current

This note explains, end to end, how the tool turns a panel's rated wattage into the power and amperage figures you see on the LED tab and in the project PDF. Every value below can be overridden per screen.

Panel wattages (current catalog)

These are the rated (peak, full-white) wattages per cabinet in the tool today. The 0.5 x 1 m cabinet exists in two rows — bare and with its captive cable loom — but both pull the same power.

Cabinet	Size (m)	Pixels	Rated power
Uniview UR Pro 0.5x1m	0.5 x 1.0	128 x 256	350 W
Uniview UR Pro 0.5x1m + cable	0.5 x 1.0	128 x 256	350 W
Uniview UR Pro 0.5x0.5m 90 deg	0.5 x 0.5	128 x 128	175 W
Uniview UR Pro 0.5x0.5m 90 deg + cable	0.5 x 0.5	128 x 128	175 W
Custom panel (fallback)	0.5 x 0.5	128 x 128	175 W

The build-up, step by step

1 Start with the panel's rated power

Every cabinet in the catalog has a rated wattage (its peak draw running full white). This is the starting point for one cabinet.

2 Scale it for brightness

You rarely run full brightness. The nominal calibration point is 5000 nits. Even at 0 nits a panel still pulls 30% of rated power (receiving cards, fans, processors never switch off) — that is the floor. Above the floor it scales linearly up to 100% at nominal.

```
brightnessFactor = 0.30 + 0.70 x (yourNits / 5000)    (capped at 1.0)
```

Example: a screen set to 2500 nits = $0.30 + 0.70 \times 0.5 = 65\%$ of rated power.

3 Add PSU overhead

Power supplies lose energy and you want a safety margin, so a percentage is added on top (default 25%, overridable per screen).

```
wattsPerCabinet = panel.power x brightnessFactor x (1 + overhead/100)
```

4 Multiply by enabled cabinets

Only cabinets that are actually powered count — disabled or blanked ones are excluded.

```
totalWatts = wattsPerCabinet x (number of enabled cabinets)
```

5 Convert watts to amps

Using the region's mains voltage (e.g. 230 V EU, 120 V US) and the power factor (default 0.95).

$\text{amps} = \text{totalWatts} / (\text{voltage} \times \text{powerFactor})$

6 Split across power chains

If you set a max-cabinets-per-chain, the tool works out how many chains you need and divides the amps evenly. A chain is flagged as overloaded when it exceeds 80% of the breaker rating (the standard de-rate).

Worked example — a 5 x 3 wall

5 cabinets wide x 3 cabinets tall = 15 cabinets, all powered. Using the Uniview UR Pro 0.5 x 1 m (350 W), at full brightness, 25% PSU overhead, 230 V EU mains, 0.95 power factor. That makes a wall 2.5 m wide x 3.0 m tall.

Step	Calculation	Result
Brightness factor	full (5000 nits) -> 1.0	1.0
Watts per cabinet	$350 \times 1.0 \times 1.25$	437.5 W
Total (Max output)	437.5×15	6,562.5 W
Current	$6562.5 / (230 \times 0.95)$	30.0 A
Average output	$6562.5 / 3$	2,187.5 W
Chains (6/chain)	$\text{ceil}(15 / 6)$	3 chains
Amps per chain	$30.0 / 3$	10.0 A

Read-out: this wall needs about 30 A total at full white — across 3 power chains that is 10 A each, comfortably under a 16 A breaker's 80% ceiling (12.8 A). For generator and heat planning, expect roughly 2.2 kW average on real content.

Dim the wall and it drops fast: the same 5 x 3 at 2500 nits uses brightness factor 0.65, so watts per cabinet = $350 \times 0.65 \times 1.25 = 284$ W, total = 4,266 W (Max), and current = 19.5 A.

Max output vs Average output

Max output = totalWatts above — peak draw, full white. Size your cable, distro and breakers off THIS number.

Average output = Max x 1/3. Real content (video, text, logos) almost never lights every pixel white, so realistic running draw is roughly a third of peak. Use it for generator fuel and heat planning — never for sizing power infrastructure.

What you can override per screen

Voltage region - Power factor (default 0.95) - PSU overhead % (default 25%) - Brightness in nits - Cabinets per power chain. Each has a sensible default but can be set per screen.

In one line

Rated cabinet watts -> dim it for brightness -> pad it for PSU losses -> multiply by cabinet count -> divide by volts to get amps -> split across chains and check against breakers.